

1 Acceptability judgements in Colloquial Singaporean English

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Abstract

Colloquial Singaporean English (CSE) includes several optional features which vary in their usage. Such features include the zero copula in copular constructions (e.g., “Tom $\emptyset$ very tall”), as well as the lack of past tense inflection in past constructions (e.g., “Yesterday, she go to the store”). In this study, we used an acceptability judgement task to estimate the effect of social factors (e.g., mother tongue language) and linguistic factors (e.g., complement type or past inflection form) on CSE speakers’ acceptability of these two classes of constructions. For copular constructions, we found that PP complements were more acceptable than NP or AP complements with a zero copula, and there were no differences in ratings between Malay and Chinese speakers. These results suggest that previously-found differences have been levelled among CSE speakers. For past tense constructions, Malay speakers showed a greater difference in ratings than Chinese speakers for sentences with versus without past tense inflection. There were also no differences among different forms of past tense inflection (suppletion, vowel change, affixation), contrasting with previous studies. These results help to complement and clarify the findings from corpus-based studies of the distribution of these optional features among CSE speakers, demonstrating the utility of using acceptability judgements to study sociolinguistic variation and language change.

Keywords: Colloquial Singaporean English, acceptability judgements, sociolinguistics, variation

Word count: 6487

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Introduction

Singapore English (SgE) is a contact variety spoken in Singapore, with substrate languages and dialects including Chinese¹ and Malay². It is spoken by most Singaporeans, the majority of whom are also bi- or multilingual, in part due to bilingual education. Students are required to learn both English and another language, termed a “mother tongue” (MT), which is typically related to the student’s ethnicity: ethnic Chinese students learn Mandarin, ethnic Malay students learn Malay, and ethnic Indian students learn an Indian language (e.g., Tamil, Hindi, Punjabi; Pakir, 1999). SgE includes a range of subvarieties, with the acrolect termed Standard Singapore English (SSE) and the basilect termed Colloquial Singapore English (CSE; also commonly known as Singlish). The continued coexistence of these subvarieties has led to a number of proposed approaches to the role of CSE in Singapore, including ‘continuum’ (Platt, 1975) and ‘diglossia’ (Gupta, 1998), *inter alia*. Perhaps the most helpful model for understanding CSE and CSE variation is Leimgruber’s indexical approach (Leimgruber, 2013), according to which linguistic variables index social positions (e.g., formality, emotion, socioeconomic aspirations) and speakers select features that appropriately indicate their desired stance rather than necessarily adopting an entire set of features associated with CSE or SSE.

¹ Our use of “Chinese” here is intentionally imprecise. When referring to an ethnic group, it denotes people of Chinese descent, but when referring to language, we use it as a term covering various Sinitic varieties spoken in Singapore, including Mandarin, Hokkien, Cantonese, and Teochew, among others, as they share relevant syntactic characteristics. Our subsequent analyses of Chinese varieties will focus on (Mainland China) Mandarin as a representative example; Singaporean varieties are mostly similar with respect to relevant matters of syntax (see Lin & Khoo, 2018; Zhu, 2016).

² Similarly, our use of “Malay” is also imprecise, referring both to a broad ethnic group, as well as a term for the Malay macrolanguage, which includes varieties spoken in Indonesia and Malaysia as well as Singapore.

Under this model, it is possible to describe CSE features without presuming that they must always co-occur within any particular utterance. Instead, one should expect variation in the usage of CSE features by speakers of SgE, and in particular, sociolinguistic factors are likely to be important predictors of variation. In order to capture the natural variation in SgE, researchers have often turned to recordings of naturalistic speech or corpus analyses as methods to understand the distribution of different features of SgE (e.g., Deterding, 2003; Ho & Platt, 1993; Lai, Tan, Wong, Loke, & Bond, 2013; Leimgruber, 2013). The ecologically valid context of these approaches has been crucial in capturing SgE as it is actually used, and has generated rich data which can be analysed along many different dimensions.

However, as is the case with any investigation using naturalistic data, these approaches have certain limitations. Firstly, because of the difficulty of collecting and processing direct recordings, naturalistic studies often employ relatively small sample sizes, limiting the generalisability of the results. This has been mitigated by the recent development of a large-scale spoken SgE corpus (Koh et al., 2019), which is likely to enable more accurate descriptions of feature distributions across SgE speakers. Nonetheless, there remains variation in the number of times any particular target phenomenon may occur within one participant's speech, and lower frequency phenomena may remain more difficult to study; this is related to the "negative data problem", which is the observation that constructions which do not appear in a corpus may not be ungrammatical, merely unattested (Bennett, 2010). Secondly, in naturalistic settings, there is no way to control for the influence of many other factors, especially pragmatic, discourse, and other contextual information. These affect the kind of language that may be appropriate for speakers to use, and in the particular case of SgE, the indexical approach suggests that speakers flexibly modulate the features they express in their speech depending on many of these background factors (Leimgruber, 2013). As such, it is difficult to characterise CSE features and their distribution while controlling for such

extraneous factors.

A complementary methodology that addresses some of these issues is the use of formal acceptability judgements. This method involves explicit collection of acceptability ratings for various constructions from native speakers, and allows for systematic capture of particular phenomena that may not be possible with naturalistic data. Additionally, formal acceptability judgements collected from a larger number of individuals permit the analysis of variation across individuals, and also decrease the potential magnitude of biases that can occur with informal acceptability judgements that are the result of the researcher’s own introspection or that represent the responses of a small number of informants (Juzek, 2015). Currently, formal acceptability judgements as data are gaining traction in linguistics research, but there are still relatively few instances of their application with respect to populations with known sociolinguistic variation, including SgE. A search for the terms “Singapore(an) English” and “acceptability judg(e)ment” in Google Scholar returned only 5 results that were true formal acceptability judgement tasks (Chan, 2023; Horsch, 2022; A. X. W. Leong, 2013; L. Li, 2015; Terassa, 2019).³

Hence, the current study sought to explore the use of formal acceptability judgements as a means of understanding the distribution of CSE features among SgE speakers. In particular, we investigated the relationship between linguistic and demographic variables as a window into sociolinguistic variation in SgE. To do so, we chose two well-attested CSE features, namely the zero copula and uninflected past tense.

³ There were a number of other studies which used participants’ *process* of generating acceptability judgements as a way to measure their linguistic attitudes and language ownership, rather than as a way to characterise SgE (Bokhorst-Heng, Alsagoff, McKay, & Rubdy, 2007; Bokhorst-Heng, Rubdy, McKay, & Alsagoff, 2010; Foo & Tan, 2019; Higgins, 2003; Rubdy, McKay, Alsagoff, & Bokhorst-Heng, 2008). There was also an additional study which technically did use an acceptability judgement task, but only with a small number of advanced linguistics students (Sato, 2013), which would again potentially limit the generalisability of the results.

Both of these features have observed optionality, and we examined how their acceptability varied with linguistic (e.g., phonetic, syntactic, or semantic) features of the stimuli, as well as with participants' language backgrounds (languages spoken and language proficiencies). In the remainder of the introduction, we provide a brief overview of the two phenomena of interest.

Copular constructions in Singaporean languages

Copular constructions in SSE. In SSE, as in many other varieties of English, the copula is *be* and is obligatory in copular constructions, as in (1).

- (1) a. Today is [_{NP} Saturday].
 b. Tom is [_{AP} very tall].
 c. Sally is [_{PP} at home].

Copular constructions in Chinese. In Chinese, the ordinary copula is 是 *shì*. Similar to other constructions in this isolating language, this does not inflect for tense, number, or person (unlike *be* in English). 是 *shì* is typically used with NP complements, as in (2); with most NP complements, the copula is obligatory.⁴

- (2) 约翰 *(是) (个) 医生。
 yuēhàn *(shì) (gè) yīshēng.
 John COP CLF doctor
 "John is a doctor."

⁴ Some NP complements seem to license constructions without a copula as informal responses to questions, such as time and price (Chang, 2009), and some varieties may also permit predicative nominals without the copula (Tang, 2001), but these tend to be exceptions and are generally less acceptable than their counterparts with an overt copula. We assume that the general case involves an obligatory copula, and that other pragmatic factors permit copula-less forms.

125 On the other hand, AP predicatives⁵ disallow the copula in Chinese, except with
 126 emphatic or focus marking, as in (3).

127 (3) 他 ∅ 很 开心。
 tā ∅ hěn kāixīn.
 3SG.M POS happy
 128 “He is happy.”

129 Locative expressions in Chinese require the word 在 *zài*, as in (4). This word does not
 130 co-occur with 是 *shì* except when the latter is used as a focus marker (Xu, 2003).

131 (4) 我 *(在) 厨房 里。
 wǒ *(zài) chúfáng lǐ.
 1SG LOC kitchen in
 132 “I am in the kitchen.”

133 We assume that 在 *zài* can be considered a locative copula, given that it is needed to
 134 define the predication structure for PPs (see Arche et al., 2019; Ross, 1991; Sharma &
 135 Rickford, 2009).⁶

136 In summary, Chinese copular constructions with an NP predicative require the
 137 copula 是 *shì*, those with an AP predicative do not have a copula, and those with a PP
 138 predicative require the locative copula 在 *zài*.

⁵ The ability of adjectives to function as clausal predicates has led to suggestions that they should be considered intransitive stative verbs rather than a separate word class (McCawley, 1992). However, distributional analyses have suggested that the adjective does indeed form a separate word class in Chinese (Arcodia, 2014; Paul, 2010), and we take this view in subsequent analyses.

⁶ There is disagreement about whether the complement of 在 *zài* is an NP or a PP, related to whether locative words such as 里 *lǐ* ‘in’ and 上 *shàng* ‘on’ are nouns (e.g., McCawley, 1992) or postpositions (e.g., Djamouri, Paul, & Whitman, 2013). We follow Y.-H. A. Li (1990) in analysing such locative words as postpositions, although the analysis would not differ significantly were they to be analysed as nouns.

Copular constructions in Malay. In prescribed rules of Standard Malay, NP predicatives can appear either with or without the copula *ialah* (Nik Safiah Karim, Farid M. Onn, Hashim Haji Musa, & Abdul Hamid Mahmood, 1989), as in (5); the examples in this section are adapted from Moeljadi, Bond, and Costa (2016) except where otherwise noted.

- (5) Budi (ialah) guru.
 Budi COP teacher
 “Budi is a teacher.”

When considering actual use of Malay, however, the copula *adalah* is also observed with NP complements (I Wayan Arka, 2013; Khairul Taufiq Abu Bakar & Nor Hashimah Jalaluddin, 2017). The two copulas do not seem to differ semantically (Moeljadi et al., 2016), although a corpus study revealed that *ialah* appears much more frequently with NP predicatives than other types of predicatives (73.8% of occurrences) as compared to *adalah* (59.3% of occurrences) (Khairul Taufiq Abu Bakar & Nor Hashimah Jalaluddin, 2017). A final note about Malay NP predicatives is that they behave differently to other copular predicatives, as they require an inchoative copula (*menjadi* ‘become’) in the future tense (unlike AP and PP predicatives), as in (6) (I Wayan Arka, 2013).

- (6) Dia akan *(menjadi) dokter.
 3SG FUT become doctor
 “He/she will become a doctor.”

While an analysis of this verb is outside the scope of the present study, it suggests that NPs have a greater tendency to occur with a copula than other types of complements in Malay (at least in some contexts).

On the other hand, prescriptive grammars for Standard Malay suggest that AP predicatives can appear either with or without the copula *adalah* (Nik Safiah Karim et

162 al., 1989), as in (7), adapted from Mustaffa (2018).

- 163 (7) Ros (adalah) merah.
 rose COP red
 164 “Roses are red”

165 Actual use shows again that, like NPs, AP predicatives can appear with *ialah*, although
 166 much less frequently (Khairul Taufiq Abu Bakar & Nor Hashimah Jalaluddin, 2017).
 167 However, speakers seem to disagree on the acceptability of either *ialah* or *adalah*
 168 appearing with AP predicatives (Moeljadi et al., 2016), thus it seems that AP
 169 predicatives strongly disfavour the presence of a copula.

170 Locative expressions in Malay can optionally take an existential copula *ada* or
 171 *berada* (Moeljadi et al., 2016), as in (8).

- 172 (8) Budi (ada/berada) di rumah.
 Budi EXIST at home
 173 “Budi is at home.”

174 Again, actual use reveals that both *adalah* and *ialah* are used with locative expressions
 175 (Khairul Taufiq Abu Bakar & Nor Hashimah Jalaluddin, 2017). This also applies for
 176 benefactive constructions, which (while not technically locative) also involve a
 177 preposition *untuk* ‘for’, as in (9).

- 178 (9) Ini (adalah/ialah) untuk Budi.
 this COP for Budi
 179 “This is for Budi.”

180 In summary, copular constructions in Malay can occur without a copula, with *adalah*, or
 181 with *ialah*. Furthermore, PP predicatives can optionally take an existential copula
 182 *(ber)ada*.

Copular constructions in CSE. CSE, like SSE, has the copula *be*. Unlike SSE, however, CSE also permits a zero copula (sometimes termed “copula omission” or “copula deletion”), exemplified in (10), adapted from Chang (2009).

- (10) a. Today $\emptyset$ [_{NP} Saturday].
 b. Tom $\emptyset$ [_{AP} very tall].
 c. Sally $\emptyset$ [_{PP} at home].

A number of studies have examined the contexts in which the zero copula is licensed (Chang, 2009; Ho & Platt, 1993; Platt, 1979; Yu, 2015). In particular, empirical data from the 1970s suggests that the distribution of the zero copula was affected by speakers’ L1 (Chinese or Malay)⁷—Platt (1979) found that Malay speakers had the zero copula most consistently with PPs, followed by APs and then NPs, while the rates for Chinese speakers were mostly equal (although slightly higher for APs). In their analysis of Platt’s data, Sharma and Rickford (2009) suggested that the difference in distributions may have been due to the influence of copular constructions in the L1s: “The greater absence of copulas in Malay may explain the higher frequencies of zero copula among Malay-medium students. Similarly, the higher rate of omission with adjectival predicatives among Mandarin and Cantonese speakers learning English may derive from its parallel absence in their first languages.” (p. 72) This suggestion arises from the observation that copulas are obligatory in Chinese for NP and PP complements (but disallowed for AP complements, except with emphatic or focus marking), whereas copulas are always optional in Malay (see Tan, 2020 for a more extensive discussion).

While this study has been seminal in CSE sociolinguistics, certain issues remain

⁷ In Platt’s study, “L1 language” was based on the language medium of the schools attended by the informants (as there were Chinese- and Malay-medium schools prior to 1987). Additionally, complements labelled “PP” only included locatives, and not temporals or other types of PP construction.

205 regarding zero copula distribution, including the issues mentioned above with using
 206 naturalistic data. Specifically, the data were based on a relatively small sample of 3
 207 Malay and 14 Chinese informants, thus the generalisability of the results is
 208 uncertain—for instance, the 100% rate of zero copula in PP complements for Malay
 209 speakers was based on only 5 tokens. Additionally, there was no way to control for other
 210 factors such as adverbials and discourse particles, which contribute pragmatic
 211 information and are known to affect the acceptability of the zero copula (Chang, 2009;
 212 Yu, 2015). Finally, and most significantly, much has changed in the sociolinguistic
 213 landscape of Singapore since 1979, when this study was published. The “Speak Mandarin
 214 Campaign” launched in 1979 encouraged Chinese Singaporeans to speak Mandarin
 215 instead of other Chinese “dialects” in order to unify the ethnic Chinese population,
 216 resulting in greater linguistic homogeneity among the Chinese. Furthermore, the
 217 introduction of the “national stream” in 1983 meant that all students were educated in
 218 English as the primary language of education by 1987, with their MT taught as their
 219 second language. Subsequent improvements in the quality and reach of education have
 220 meant that most current Singaporeans are effectively bilingual, and many would consider
 221 English to be the primary language of their household (Woon, 2018; Wu, O’Brien, Styles,
 222 & Chen, 2020); thus, “L1” is no longer a helpful term to designate the substrate languages
 223 of CSE, and “MT” is more relevant as a label for the language(s) other than English in
 224 which Singaporeans are educated, whether or not this is actually their “first” or
 225 “household” language. These changes in the linguistic landscape suggest that more
 226 research is necessary to understand copular constructions in contemporary SgE.

227 More recent characterisations of CSE copular constructions have taken a more
 228 theoretical approach, focusing on possible origins of copula optionality (Ansaldi, 2010;
 229 Chang, 2009) and syntactic principles underlying constructional variation (Yu, 2015).
 230 While these have provided detailed syntactic and sociolinguistic analyses of CSE copular
 231 constructions, they are limited in relying solely on the linguist’s intuitions, or possibly

Table 1

Our summary of relative acceptabilities of zero copular constructions, based on prior grammatical and empirical work.

Language	Relative acceptability of zero copula
Chinese	AP $\gg$ NP = PP
Malay	AP $>$ PP $>$ NP
CSE–L1 Chinese (1970s)	AP $>$ NP $\approx$ PP
CSE–L1 Malay (1970s)	PP $?>$ AP $>$ NP

those of a handful of informants. Issues with this approach are immediately apparent when examining examples given in descriptions of CSE copular constructions. For example, (11) is given by both Chang (2009, p. 30) and Yu (2015, p. 11); Chang considers it to be perfectly acceptable, while Yu marks it with ‘*/?’, indicating that he considers it at least somewhat unacceptable.

(11) Tom $\emptyset$ clever.

Hence, a more thorough and nuanced description of the graded distribution of zero copula acceptability is needed to understand its standing as a feature of SgE.

Synthesising the descriptions of copular constructions in Chinese, Malay, and CSE, we show the relative acceptability for zero copular constructions by predicative type in Table 1.

Past tense constructions in Singaporean languages

The presence of verbal inflection is another sociolinguistic variable in SgE. For example, past tense inflection is obligatory in SSE, as in (12).

- 246 (12) a. Yesterday, she went to the store.
 247 b. He took the bus home last week.
 248 c. They packed up this morning.

249 In contrast to English, verbs are not marked for tense in Chinese (Liu, 2015) or Malay
 250 (Winstedt & Wilkinson, 1913), as in (13) and (14) respectively.

- 251 (13) 他 上 个 星期 搭 巴士 回家。
 tā shàng gè xīngqī dā bāshì huíjiā.
 3SG.M previous CLF week take bus return.home
 252 “He took the bus home last week.”

- 253 (14) Dia menaiki bas pulang minggu lepas.
 3SG take bus go.home week previous
 254 “He/she took the bus home last week.”

255 CSE allows optional past tense marking. Thus, the examples in (15) are permissible;
 256 these examples involve the non-finite form of the verb instead of the corresponding verb
 257 with past inflection.

- 258 (15) a. Yesterday, she go to the store.
 259 b. He take the bus home last week.
 260 c. They pack up this morning.

261 This optionality is highlighted by variable past tense marking even within a single
 262 utterance, as shown in (16).

- 263 (16) a. We went in, take half an hour to come out. (Fong, 2004)
 264 b. They never say the date, he didn’t state the duration, right. (Leimgruber,
 265 2013)

More generally, CSE speakers exhibit substantial intra- and inter-utterance variability in past tense inflection (Deterding, 2007; Leimgruber, 2013). However, this variability does not appear to be entirely random or arbitrary; instead, there may be patterns in the likelihood of past tense marking (Cruz-Ferreira, 2005; Deterding, 2007; Fong, 2004; Gut, 2009; Ho & Platt, 1993). Empirical data (from Cruz-Ferreira, 2005; Gut, 2009; Ho & Platt, 1993; W. Leong, 2021 i.a.) have suggested that there are patterns in the likelihood of past tense marking, such that verbs whose past form ends in a consonant cluster (e.g., *picked*) are less likely to be inflected, whereas verbs whose past forms involve a vowel change (e.g., *fell*) are more likely to be inflected. This pattern of results may be due to a number of (non-exclusive) factors. First, final consonant clusters tend to be reduced in SgE (Cruz-Ferreira, 2005; Deterding, 2007; Lim, 2004); thus, past forms ending in consonant clusters may simply be reduced to the stem form without the final [t]/[d]. Notably, however, omission of final [t]/[d] appears to be more frequent in other lexical words in comparison to the regular past *-ed* morpheme (Cruz-Ferreira, 2005; Deterding, 2007). Second, irregular past forms (including those involving a vowel change) tend to be more frequent (Gut, 2009; Lieberman, Michel, Jackson, Tang, & Nowak, 2007), which may help to preserve the marked form. Nonetheless, even accounting for morphophonological effects, individual verbs still vary substantially in their likelihoods of being inflected for past tense (Ho & Platt, 1993).

One important gap in the current research relates to the role of MT in past tense inflection. The study by W. Leong (2021) suggested that there was no significant effect of whether participants reported primarily speaking English or a non-English variety at home on production of inflected past forms. However, this approach collapsed speakers of different MTs, and therefore could not determine the effect of MT. In contrast, a study by Wong (2012) on Malaysian learners of L2 English suggested that Malay speakers were more likely to endorse non-standard tense and agreement morphological marking than Chinese speakers, although they did not test for past tense marking. Further research is

needed to determine whether this variation also holds for speakers of SgE with different MTs.

Methods

Participants

83 participants aged 18–25 (49 male, 33 female, 1 other) participated in the main study. All participants were native CSE speakers. As their MT, 62 spoke Chinese, 18 spoke Malay, and 3 spoke another language (English or Tamil). Only the Chinese- and Malay-speaking participants were used for the remainder of the analyses.

Out of the 80 included participants, 73 either attended, were attending, or were about to attend university, of whom 58 were Chinese-speaking and 15 were Malay-speaking. These proportions were not significantly different at $\alpha = .05$ ($\chi^2 = 1.82$, $p = .177$). Both the MT and education level distributions were similar to those of the overall 18–25 population in Singapore (Department of Statistics, Ministry of Trade and Industry, Republic of Singapore, 2021).

40 participants indicated that they could speak at least one variety other than English and their MT. Of these, 23 of the Chinese speakers could speak another Chinese variety (Cantonese, Hokkien, Teochew, or Henghua), and 8 of the Chinese speakers could speak Malay. 3 Malay speakers could also speak Mandarin. These data were not excluded as these multilingual speakers uniformly rated themselves as being more proficient and having more formal training in their MT than their third language.

Task

Participants completed a questionnaire online using Qualtrics, in which they evaluated sentences on a 1–5 scale based on their naturalness (see Appendix). They were told not to consider grammar rules they may have learnt in school, but to focus on their

sense of what would be appropriate in ordinary relaxed conversation. Participants were also reminded that they would not be evaluated on the basis of their responses.

The first block comprised 10 practice sentences, including 5 sentences in SSE and 5 in CSE. This block allowed participants to become familiar with the task, and also primed them to interpret sentences in a CSE context.

Subsequently, participants completed three blocks of 24 questions each. Each block contained 12 copula test sentences, 10 past tense test sentences, and 2 benchmark sentences.⁸ In particular, the 12 copula test sentences included 4 sentences for each complement type (NP, AP, PP); these all involved a 3SG subject to control for the form of the copula (i.e., *is*). The 10 past tense test sentences included 2 sentences of each past tense type (suppletion, vowel change, [t]/[d] affixation, [ɪd] affixation, vowel change + [t]/[d] appending); they also all included a past temporal marker (e.g., *yesterday*), providing a temporal discourse frame that would both signal that the event took place in the past, and license uninflected past forms (Gut, 2009; Ho, 2003). The test sentences had both SSE and CSE versions (i.e., with and without the copula, or with and without past tense marking respectively), such that each participant would be presented with only one version of each test sentence. Participants rated an equal number of SSE and CSE sentences. Which sentences were presented in CSE was counterbalanced across participants. One low and one high benchmark sentence were also included to verify that participants understood the task. The full list of sentences can be found in the Appendix.

After completing the three test blocks, participants responded to a set of demographic questions, including information about age, gender, education, MT language proficiency, and duration of living in Singapore. The full list of demographic questions can be found in the Appendix.

⁸ The copula and past tense test sentences mutually served as fillers, and thus no additional filler sentences were included. No participant reported identifying both classes of test sentences, suggesting that this method of blinding was sufficient.

All participants gave informed consent before taking part in the study.

Results

Copular constructions

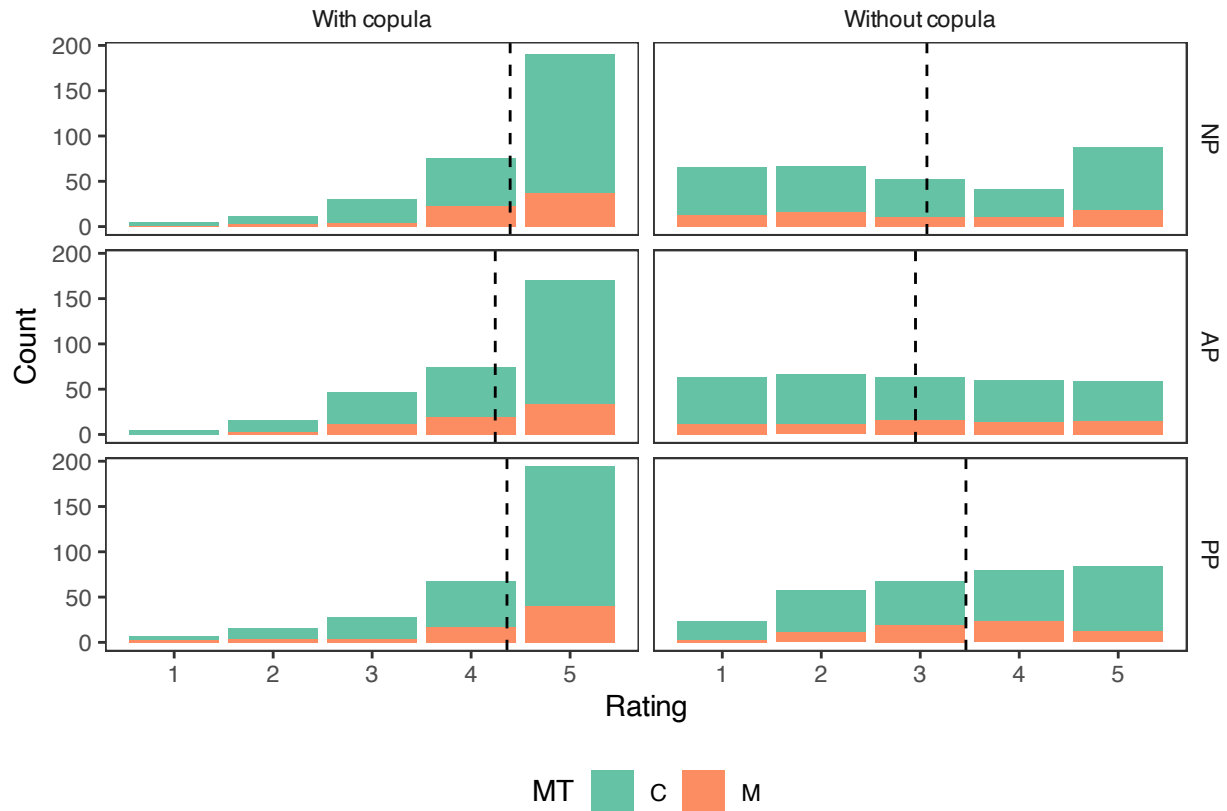


Figure 1. Distributions of ratings for copular constructions by complement type, sentence version, and MT. Black dashed lines indicate means. C: Chinese; M: Malay.

Participants' ratings for items involving copular constructions are shown in Figure 1. We conducted a linear regression with rating as the outcome variable, and complement type, sentence version, and MT as fixed effects with sum contrasts, along with all interactions. We also included participant and item as random effects. The full model specification is as follows: $\text{rating} \sim \text{comp_type} * \text{sent_version} * \text{mt} + (1 \mid \text{ppt_id})$

349 + (1 | item_id).⁹

350 Model results suggested that there was an unsurprising main effect of sentence
 351 version ($b = 1.16$, $p < .001$), with sentences including the copula having higher mean
 352 ratings (4.33 ± 0.10) than sentences without the copula (3.17 ± 0.10).¹⁰

353 There was no main effect of MT ($b = 0.05$, $p = .733$), suggesting that mean ratings
 354 did not differ between Chinese speakers (3.77 ± 0.09) and Malay speakers (3.72 ± 0.09).

355 However, there were interaction effects between complement type and question
 356 type, as shown in Figure 2. Post-hoc pairwise comparisons using the Tukey method
 357 revealed that there were no significant pairwise contrasts for sentences including the
 358 copula (all $p > .20$). However, there were significant pairwise contrasts for sentences
 359 without the copula. Sentences with a PP complement (3.49 ± 0.15) had higher ratings
 360 than sentences with an NP complement (3.05 ± 0.15 , $b = 0.54$, $p = .012$) and also had
 361 higher ratings than sentences with an AP complement (2.96 ± 0.15 , $b = 0.44$, $p = .048$).
 362 Ratings for sentences with an NP complement were not significantly different from
 363 sentences with an AP complement ($b = 0.10$, $p = .851$).

364 Surprisingly, there were no interaction effects involving MT. There was no
 365 interaction between language and complement type ($p > .20$), or language and sentence
 366 version ($p = .367$), or language and both complement type and sentence version ($p > .40$).

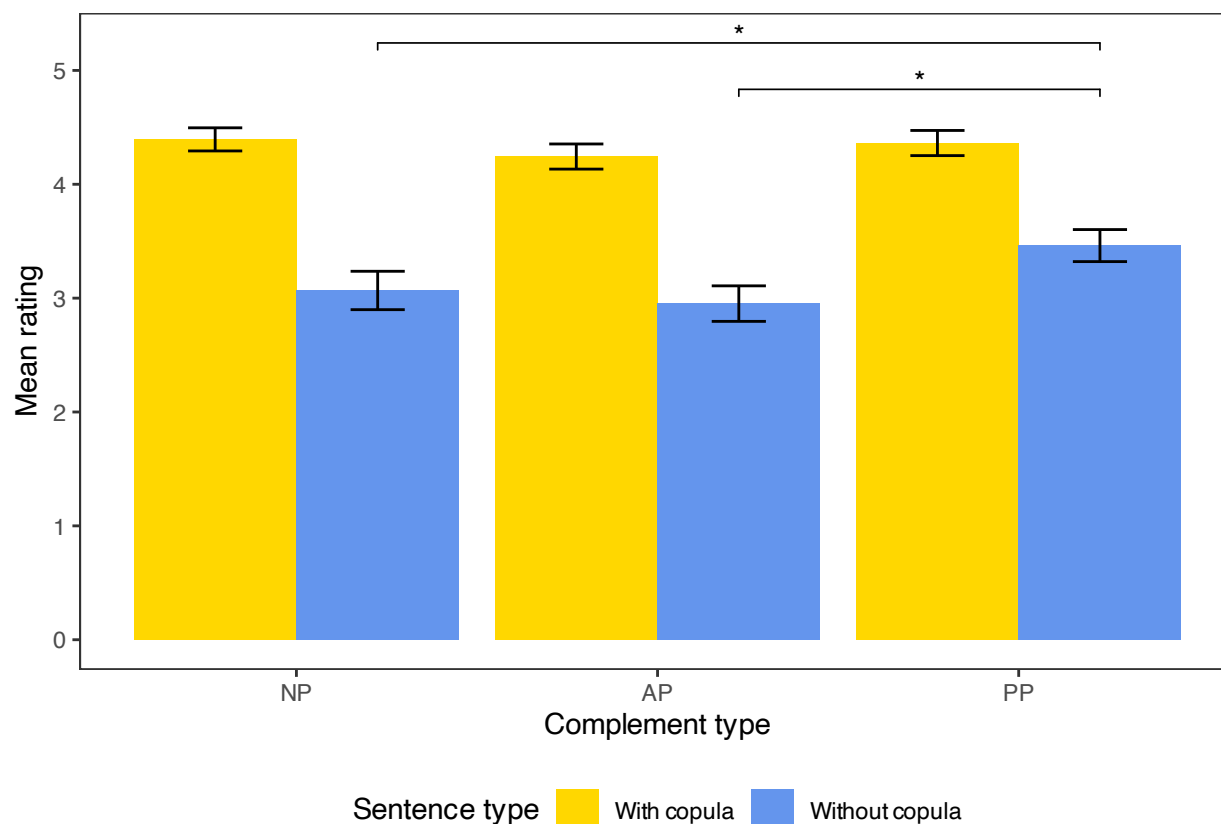


Figure 2. Mean ratings for copular constructions by complement type and sentence version.

*: $p < .05$, **: $p < .01$.

Past tense marking—past marking type

Participants' ratings for items involving past tense constructions are shown in Figure 3. We conducted a linear regression with rating as the outcome variable, and past marking type, sentence version, and MT as fixed effects with sum contrasts, along with all interactions. We also included participant and item as random effects. The full model

⁹ A Shapiro–Wilk test revealed that residuals of the linear regression were not normally distributed ($W = 0.99$, $p < .001$). However, visual observation of the Q–Q plot suggested that the deviation was minor and only occurred near the top tail of the distribution, which was expected due to negative skew (-0.72) associated with a ceiling effect. As such, linear regression remained the analysis of choice.

¹⁰ Reported values indicate estimated marginal means and their standard errors.

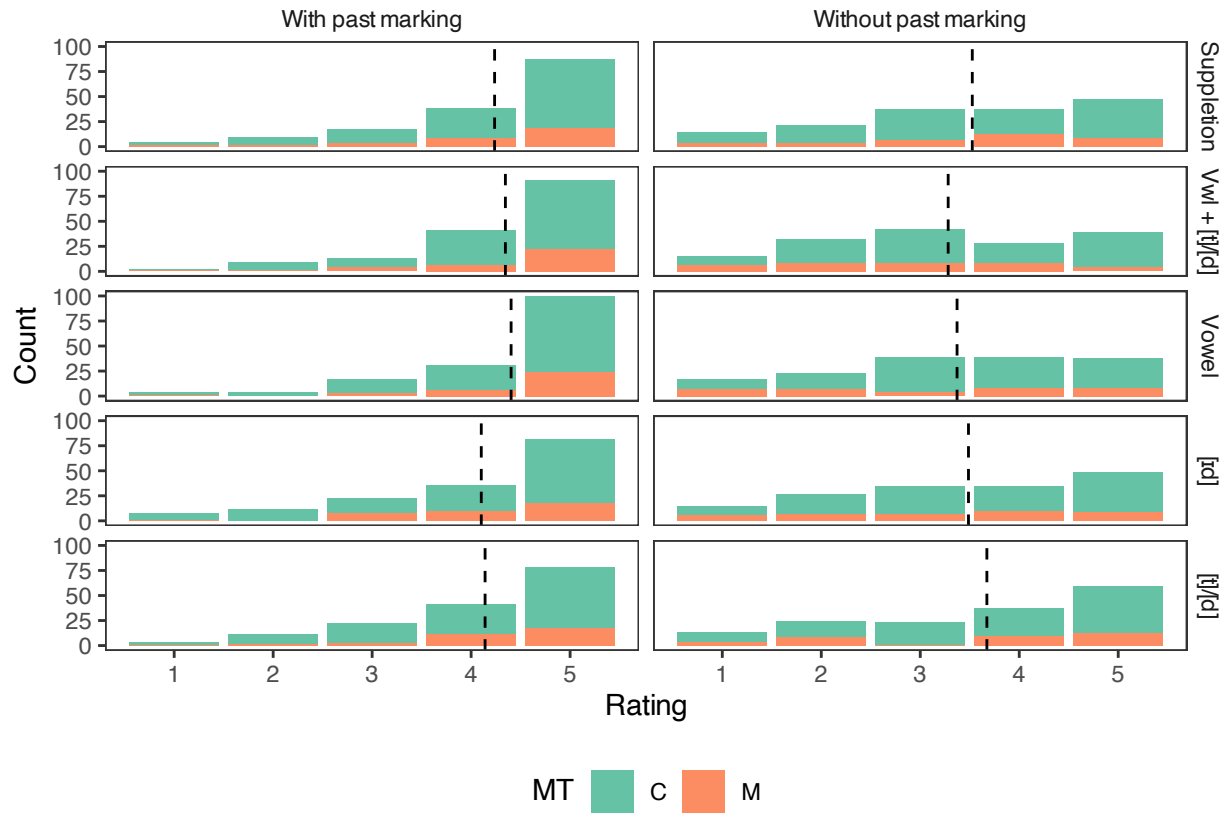


Figure 3. Distributions of ratings for past tense constructions by past tense marker, sentence version, and MT. Black dashed lines indicate means. C: Chinese; M: Malay.

specification is as follows: `rating ~ past_type * sent_version * mt + (1 | ppt_id)`
`+ (1 | item_id)`.¹¹

Model results suggested that there was again a main effect of sentence version ($b = 0.84, p < .001$), with sentences including the copula having higher mean ratings (4.24 ± 0.11) than sentences without the copula (3.40 ± 0.11).

¹¹ Similar to the ratings for copular constructions, a Shapiro–Wilk test revealed that residuals of the linear regression were not normally distributed ($W = 0.99, p < .001$). Again, visual observation of the Q–Q plot suggested that the deviation was minor and only occurred near the top tail of the distribution, which was expected due to negative skew (-0.81) associated with a ceiling effect. As such, linear regression remained the analysis of choice.

There was again no main effect of MT ($b = 0.15$, $p = .380$), suggesting that mean ratings did not differ between Chinese speakers (3.89 ± 0.10) and Malay speakers (3.74 ± 0.10).

In contrast with the pattern of results for copular constructions, there were no significant interaction effects between past marking type and sentence version (all $p > .08$). There were also no significant interaction effects between past marking type and both question type and MT (all $p > .40$).

Conversely, there was a significant interaction effect between sentence version and MT ($b = -0.34$, $p = .005$), as shown in Figure 4. Post-hoc pairwise comparisons using the Tukey method did not reveal any simple effects of MT (all $p > .07$), suggesting that Chinese and Malay speakers did not differ in their ratings for either sentence version, although they differed in the *difference* in ratings between sentences with and without past tense marking.

Past tense marking—event aspect

As past marking type was not a significant predictor of ratings for sentences involving the past tense construction, we conducted an exploratory analysis to determine whether there were other possible predictors that would explain the variation observed in participants' responses to different past tense items. One candidate was the aspect of the event semantics, which was noted by Ho and Platt (1993) also to influence the proportion of past marking. Specifically, they found that verbs which were telic showed the highest proportion of past marking, followed by stative verbs, followed by atelic verbs.¹² To explore the role of event semantics in past tense marking, we evaluated each sentence in

¹² Note that in Ho and Platt (1993), they used the terms “punctual” and “non-punctual”, but in fact the criterion used to distinguish between these two categories is explicitly characterised as the former category having the feature [+telic] or [+completive].

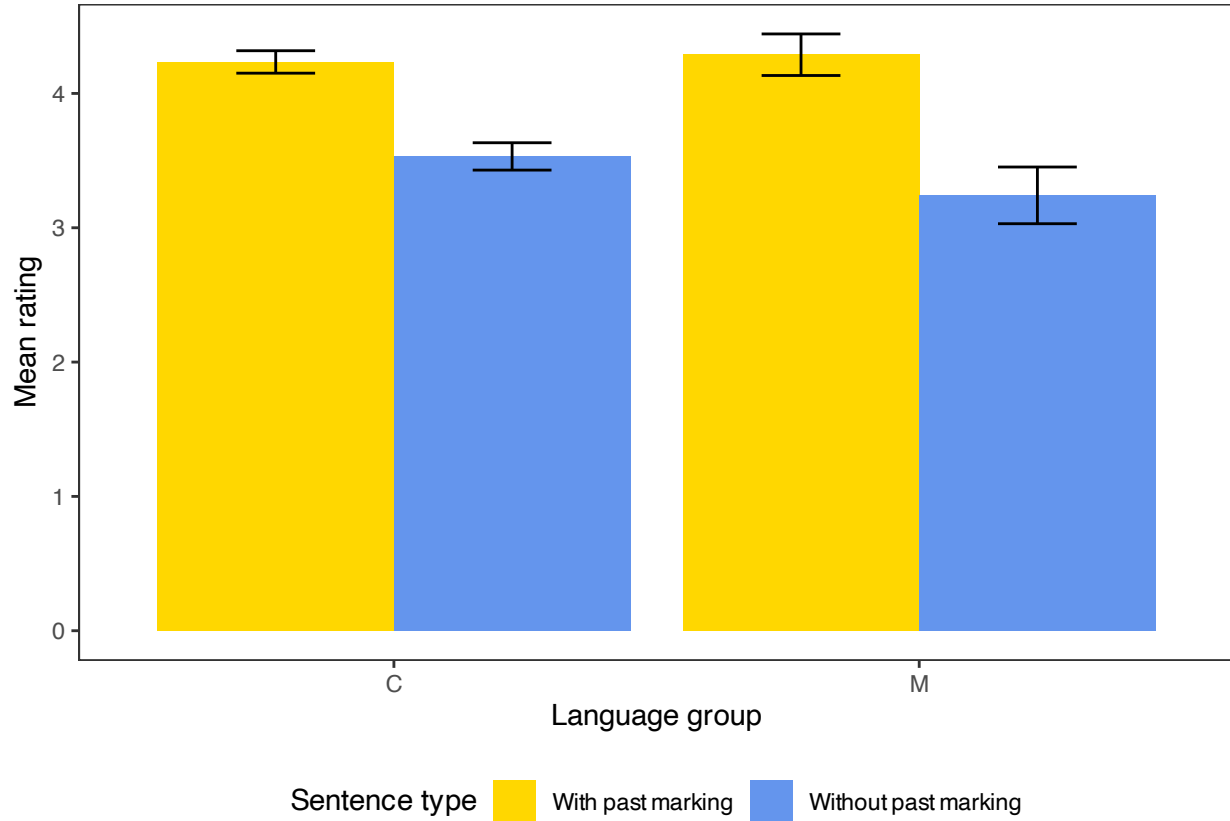


Figure 4. Mean ratings for past tense constructions by MT and sentence version. C: Chinese; M: Malay.

terms of the aspectual features of the event described: dynamic or stative, durative or punctual, and telic or atelic (Dowty, 1979; Olsen, 1994; Vendler, 1957). Using these features, we classified each sentence into aspectual classes: accomplishment ([+dynamic, +durative, +telic]), achievement ([+dynamic, -durative, +telic]), activity ([+dynamic, +durative, -telic]), or state ([-dynamic, +durative, -telic]).¹³ It is important to emphasise that this set of analyses is exploratory and post-hoc; as such, the number and order of presentation of sentences of different aspectual classes was not balanced within participants. A histogram of the number of sentences of each aspectual class is shown in

¹³ Other classes such as semelfactives ([+dynamic, -durative, -telic]) were not present among the sentences in our questionnaire.

407 Figure 5.

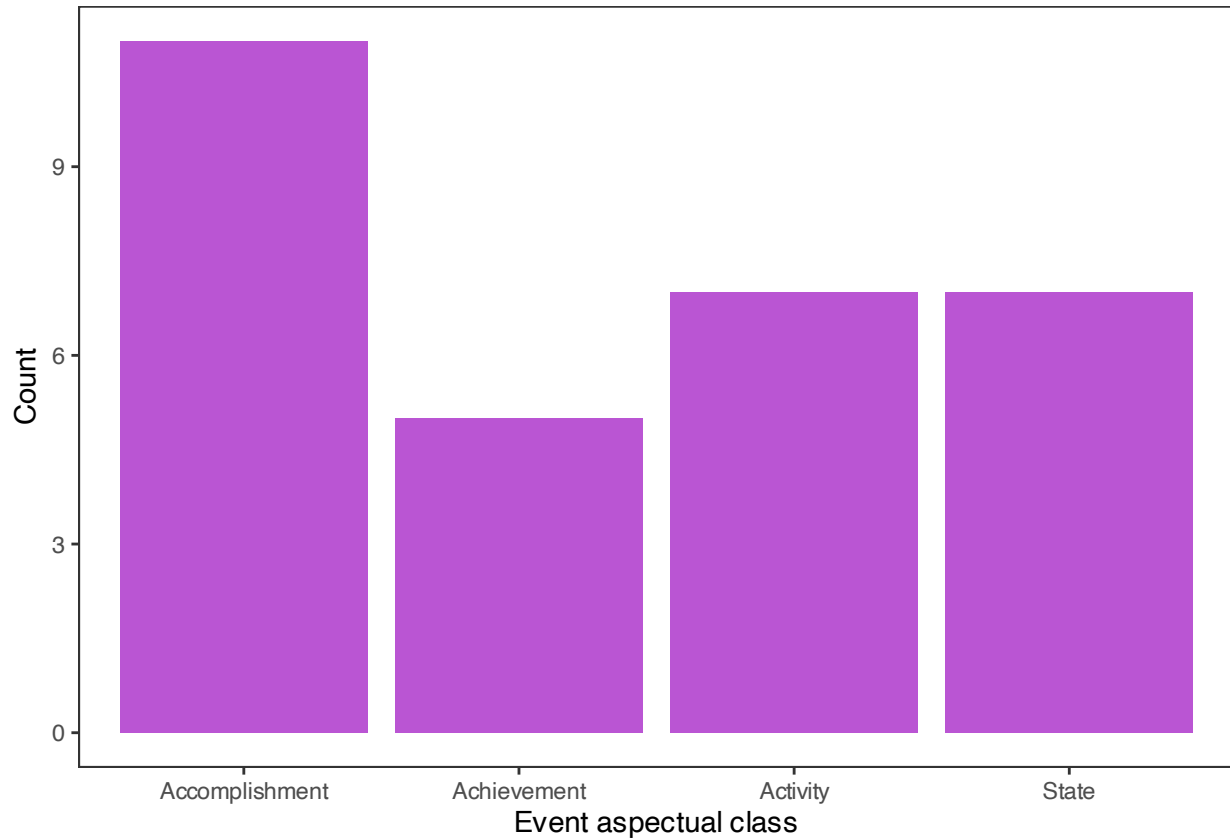


Figure 5. Histogram of aspectual classes of past tense constructions.

408 Participants' ratings for past tense constructions by aspectual class are shown in
 409 Figure 6. We conducted a linear regression with rating as the outcome variable, and
 410 aspectual class, sentence version, and MT as fixed effects with sum contrasts, along with
 411 all interactions. We also included participant and item as random effects. The full model
 412 specification is as follows: `rating ~ aspectual_class * sent_version * mt + (1 |`
 413 `ppt_id) + (1 | item_id)`.

414 Model results suggested that there was again a main effect of sentence version ($b =$
 415 0.89 , $p < .001$), and an interaction effect between sentence version and MT ($b = -0.38$,
 416 $p = .004$). Additionally, there was an interaction effect among sentence version, MT, and
 417 event aspectual class, as shown in Figure 7. Post-hoc pairwise comparisons using the

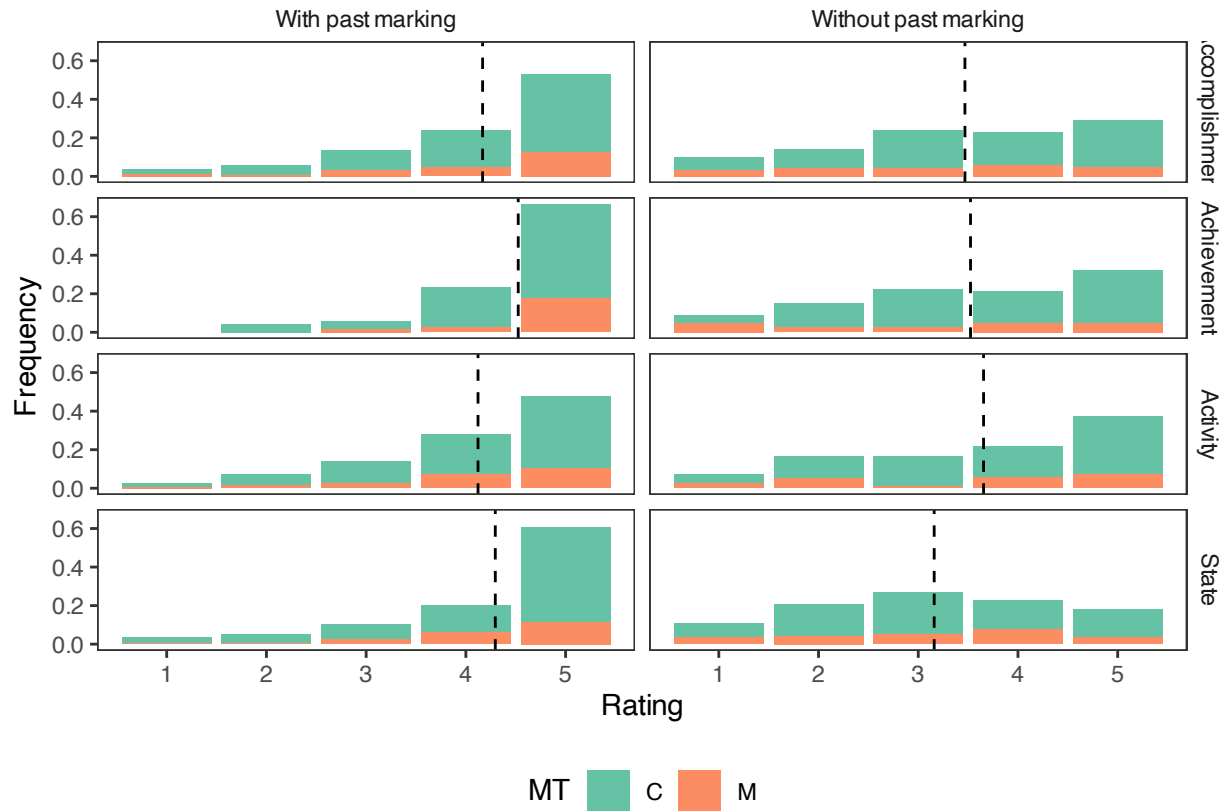


Figure 6. Distributions of ratings for past tense constructions by event aspect, sentence version, and MT. Black dashed lines indicate means. C: Chinese; M: Malay.

Tukey method revealed that there were no significant pairwise contrasts for sentences with past tense marking (all $p > .40$). For sentences without past tense marking, there were no significant pairwise contrasts for Malay MT speakers (all $p > .70$). However, for Chinese MT speakers, sentences that expressed unmarked past activities (3.74 ± 0.16) had higher ratings than sentences that expressed unmarked past states (3.13 ± 0.17 , $b = 0.61$, $p = .031$).

Individual differences

To better understand the observed variation in responses, we conducted two further exploratory analyses. The first analysis involved understanding the relationship in

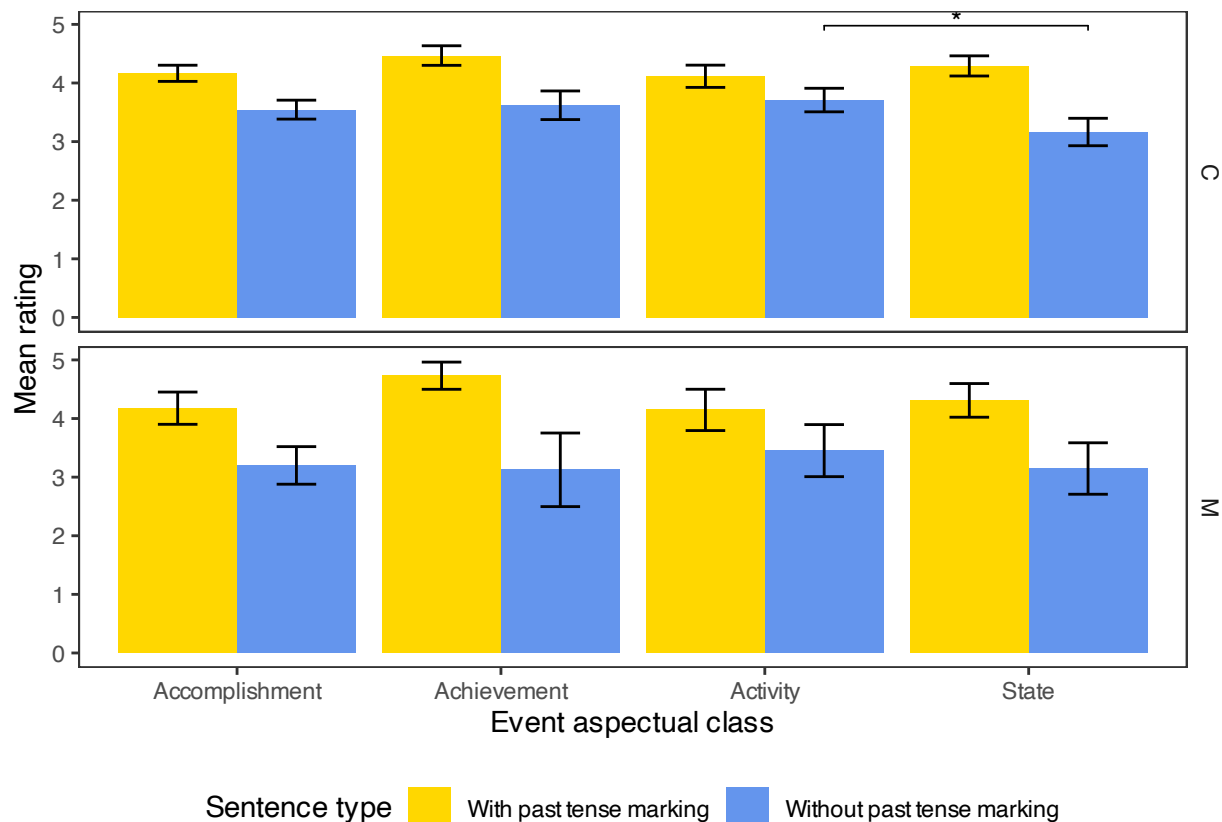


Figure 7. Mean ratings for past tense marking by event aspectual class and sentence version. *: $p < .05$.

participants' responses to the two types of constructions studied: Were participants who rated CSE copular constructions more highly also more likely to rate CSE past tense constructions more highly? We calculated the mean and 95% confidence intervals for both types of CSE constructions for each participant, shown in Figure 8.

The correlation in means between the two types of CSE constructions was moderate, $r = .65$, suggesting that participants sometimes (but not always) endorsed both types of CSE constructions with similar ratings. Additionally, many of the points lay above the $y = x$ line, suggesting that participants generally rated the CSE past tense constructions higher than the CSE copular constructions; this aligns with findings from Leimgruber (2013), who identified lack of verbal inflection as being more common than zero copula.

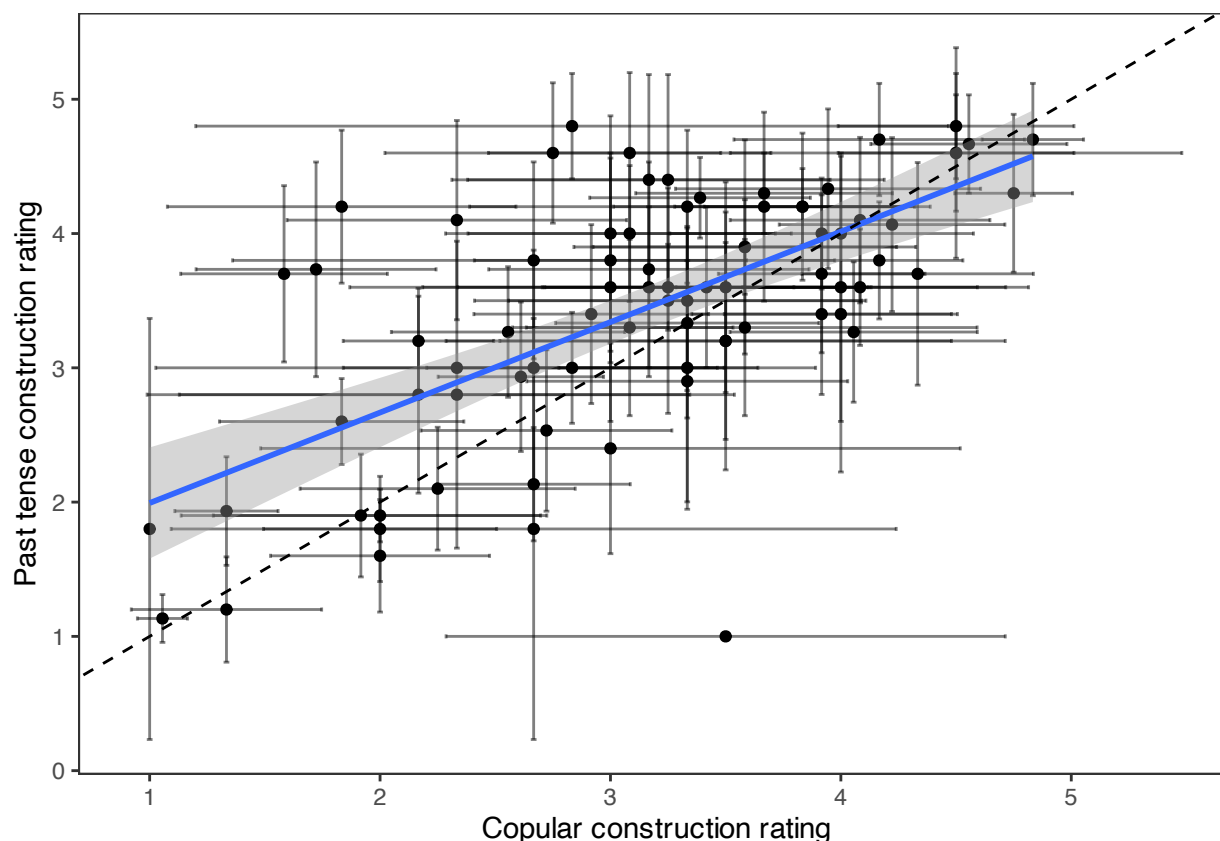


Figure 8. Ratings for copula and past tense CSE constructions by participant. Error bars indicate 95% confidence intervals. Blue line and shaded area indicate the line of best fit and 95% confidence intervals.

The second analysis studied the relationship between participants' ratings and their self-assessed MT language proficiency. A histogram of participants' MT proficiencies is shown in Figure 9.

As the sample size did not support running a full statistical model with all levels of interaction, we omitted copula or past tense type as predictors, and instead ran a single joint model on both constructions. We conducted a linear regression with rating as the outcome variable, and construction, sentence version, MT, and MT proficiency as fixed effects, along with all interactions. Construction, sentence version, and MT were coded with sum contrasts, and MT proficiency was standardised. We also included participant

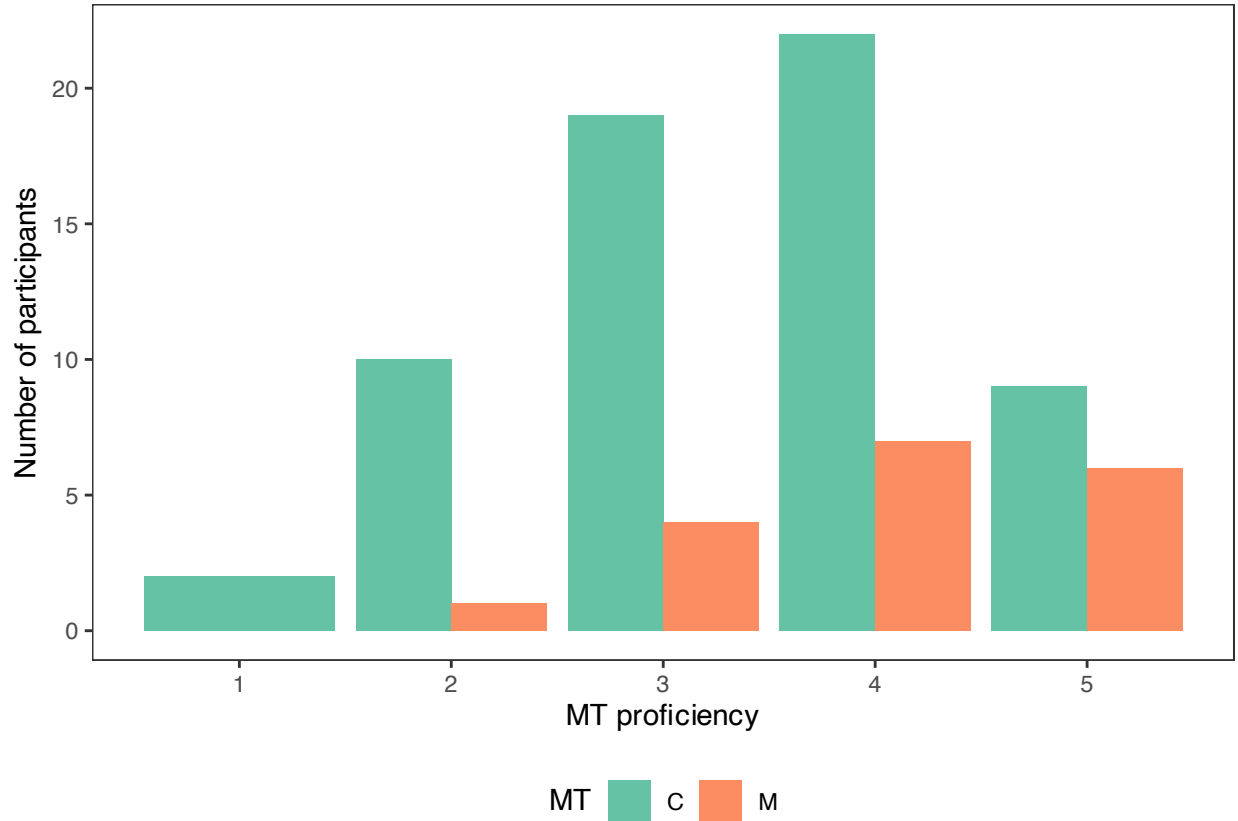


Figure 9. Histogram of participants' self-assessed MT language proficiency by MT. C: Chinese; M: Malay.

and item as random effects. The full model specification is as follows: `rating ~`
`construction * sent_version * mt * proficiency + (1 | ppt_id) + (1 |`
`item_id)`.

Participants' ratings by their self-assessed MT proficiency are shown in Figure 10. We focus on model results specifically related to MT proficiency, as other results were similar to results reported in the previous sections. There was an interaction effect between sentence version and MT proficiency ($b = -0.20$, $p < .001$), with the estimated marginal trend of MT proficiency being negative for SSE sentences ($b = -0.10$) and positive for CSE sentences ($b = 0.10$), although neither reached significance (all $p > .20$).

There was also an interaction effect among sentence version, MT, and MT

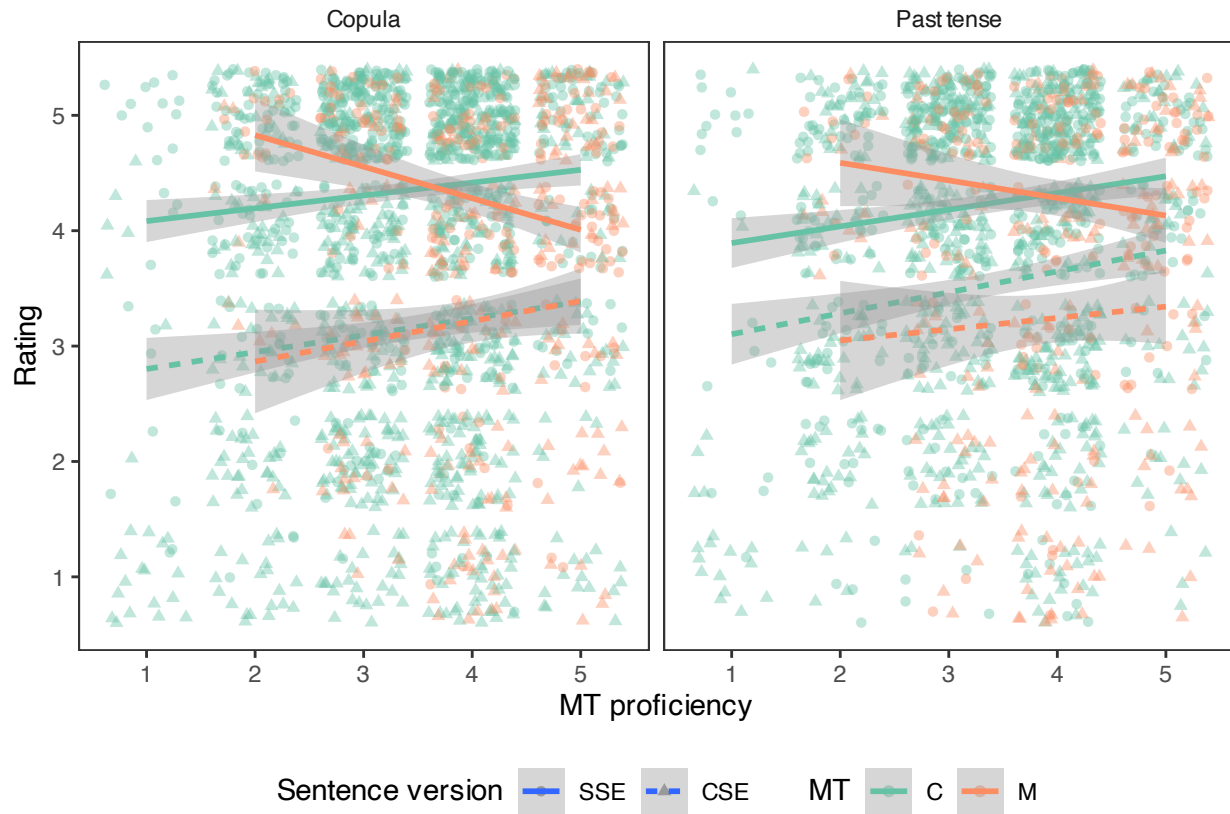


Figure 10. Ratings for copula and past tense constructions by participants' self-assessed MT language proficiency, sentence version, and MT. C: Chinese; M: Malay.

proficiency ($b = 0.36, p < .001$). For Chinese MT speakers, the estimated marginal trend of MT proficiency was positive for both SSE ($b = 0.11$) and CSE ($b = 0.13$) sentences, whereas for Malay MT speakers, the estimated marginal trend of MT proficiency was negative for SSE sentences ($b = -0.31$) but positive for CSE sentences ($b = 0.07$). Only the negative SSE trend for Malay MT speakers reached significance ($p = .043$).

In summary, as MT proficiency increased, ratings for CSE sentences also increased for both Chinese and Malay speakers. However, ratings for SSE sentences increased for Chinese speakers but *decreased* for Malay speakers as MT proficiency increased.

Discussion

Using acceptability judgements from Chinese and Malay MT speakers, we found that ratings for different sentences varied across constructions, complement or past tense type, and individual speakers. Specifically, for copular constructions, ratings for PP complements were higher than ratings for NP or AP complements when the copula was absent. These results suggest that MT has no effect on acceptability ratings of copular constructions by CSE speakers. This contrasts with earlier work from Platt (1979), which suggests that there were greater differences in the copula use of Chinese- and Malay-speaking Singaporeans in the 1970s. In comparison, the data in the present study suggest that much of the difference has been eliminated in contemporary CSE, indicating levelling of variability (see also Kalaivanan, Sumartono, & Tan, 2020). Thus, it is likely that continued interactions between Chinese and Malay speakers, as well as increased standardisation and amount of education, have resulted in the levelling of syntactic variation between the two groups. Focusing on the pattern of acceptability ratings across different complement types, it seems that PP complements are most acceptable without a copula, followed by NP and then AP (which may not differ significantly from each other). One possible (but highly speculative) explanation for this pattern is that the preposition in the PP is gaining some ‘copular’ function—i.e., linking the subject with a (spatiotemporal) position—perhaps due to the influence of the Chinese locative copula 在 *zài*.

In contrast, there were no differences across past tense types in CSE speakers. This again differs from earlier work from Ho and Platt (1993), who found that past tense forms involving a vowel change were more likely to be produced than past tense forms involving [ɪd] affixation or vowel change and [t]/[d] appending, and past tense forms involving only [t]/[d] affixation were almost never produced. One explanation for this finding is that the variation in past tense marking is strongly phonetic in nature, and therefore the

translation into the visual presentation modality was sufficient to level any differences in endorsement of unmarked past tense forms (but see Ho, 2003; W. Leong, 2021 for uninflected past tense forms in written SgE). A natural extension for further research would thus be to collect acceptability judgements using stimuli presented aurally, which may reveal more nuanced variation that was not captured by the present questionnaire.

Interestingly, for past tense constructions, we observed an interaction between sentence version and MT, such that Malay speakers had larger differences in ratings between SSE and CSE sentences than Chinese speakers. This effect was not observed by Ho and Platt (1993), who did not analyse past tense marking as a function of MT. As we did not find any simple main effects related to this interaction effect, it is difficult to pinpoint the source of this disparity. However, exploratory analyses suggested that Chinese speakers rated unmarked activities higher than unmarked states, which was in the same direction as the pattern of production found by Ho and Platt (1993). The present study did not have a full combinatorial set of sentences covering different past marking types and event aspectual classes, and a more comprehensive survey may be required to better characterise the acceptability of unmarked past tense constructions in CSE.

Our questionnaire also demonstrated variation across individual respondents. Participants' responses to copular constructions were moderately correlated with their responses to past tense constructions, suggesting that CSE features are often bundled together, but not tightly—speakers may sometimes accept or even adopt different features to differing extents. Speakers' responses were also related to their self-assessed MT proficiency in an interesting pattern of responses. As may be expected, speakers with greater MT proficiency tended to rate CSE sentences more highly, as their MT language may exert a larger influence on their CSE. However, Chinese and Malay speakers showed opposite patterns of responses for SSE sentences, as Malay speakers with greater MT proficiency rated SSE sentences more poorly, whereas Chinese speakers tended to rate

517 them more highly. These results are perhaps due to speakers' differing SSE proficiencies,
518 which may have complex interactions with speakers' MT proficiencies; a general survey of
519 Singaporeans' self-rated proficiencies across all of the languages that they speak may be
520 insightful in this regard.

521 **Language change in CSE**

522 In the absence of longitudinal studies of these constructions in the CSE of speakers
523 with different MTs, comparison of our results with data from research in the 20th century
524 can provide fresh insights into the grammar of CSE and how it has changed. As noted
525 earlier, the data in Platt (1979) provide important information about the CSE of over 40
526 years ago. Despite the previously noted shortcomings of Platt's study and the different
527 methodology employed in his work and ours, it is nonetheless illuminating to compare
528 the two and consider how and why CSE has changed since the 1970s.

529 Beginning with past tense marking, our findings indicate no difference relating to
530 MT with respect to the form of past tense marking or its absence. This is evidence of
531 convergence by all speakers on a consistent set of forms, even if Malay MT speakers'
532 ratings differed more than those of Chinese MT speakers. Similar to findings from
533 Deterding (2005) and Kalaivanan et al. (2020) on phonology, our results provide support
534 from morphosyntax for the claim that CSE is close to or has already entered the fourth
535 stage (of five) in Schneider's (2007) Dynamic Model of the evolution of postcolonial
536 Englishes. Schneider identifies this stage as endonormative stabilization, which is
537 characterised linguistically by homogeneity of features and stabilization.

538 As with past tense marking, copular constructions displayed convergence on a single
539 pattern of relative acceptability for zero copular constructions that showed no effect of
540 MT. This finding is further evidence of endonormative stabilization in the morphosyntax
541 of CSE. Interestingly, the levelling that has occurred has resulted in a pattern of relative

acceptability for zero copular constructions that is not consistent with that of either of the MT languages or the pattern attested in the CSE of L1 Malay or L1 Chinese speakers in the 1970s (Platt, 1979); see Table 1. The high ranking of PP with zero copula is possibly consistent between CSE-L1 Malay in the 1970s (PP? > AP > NP) and CSE in the 2020s (PP > NP > AP), but beyond that there are no shared patterns in the ordering. Convergence is not towards any existing pattern. It is not clear what might have driven the change to the resulting hierarchy of relative acceptability; no obvious explanation based on functional and/or categorial differences is available. This result is a matter that would benefit from further investigation. Comparison indicates that endonormative stabilization involving morphosyntactic features such as past tense marking and copular constructions may progress at different rates. Which features are distinct in this respect and how/why the rates may differ are similarly interesting questions for future research to address.

Directions for future work

One inadvertent limitation of the main study is participant distribution. The participant pool was imbalanced with regard to MT, and there was also a high proportion of university students. The latter observation is particularly important as individuals of lower socioeconomic status tend to exhibit more basilectal features in their language use (Platt, 1979), which may have been relevant in this study. Nonetheless, the relative proportions of Malay-speaking and Chinese-speaking respondents are comparable to those of ethnic Malay and Chinese Singaporeans aged 18–25, and a vast majority of this age range has attended, is attending, or is about to attend university. Furthermore, the participants were recruited by word of mouth, without specific soliciting of participants from a particular MT or socioeconomic background (to avoid demand characteristics). Future studies using a different recruitment method and a stratified sampling approach would help to alleviate this issue, and would also open the possibility

of studying the judgements of speakers of other MTs (e.g., Indian languages).

Another issue that arose in the course of data analysis is that SSE sentences may have exhibited a ceiling effect. All three complement types had a mean rating of > 4 and a skew of < -1 , suggesting that acceptability for these sentences may be almost at maximum, resulting in difficult result interpretation due to limited variation. However, this result is unsurprising given that SSE is the primary medium of education in schools, and is often considered the prestigious or ‘correct’ variety, thus there is little reason for participants to score SSE sentences poorly (other than possible ‘stiltedness’ if they felt they were unlikely to use such sentences in ordinary CSE speech). Nonetheless, future research in this area may employ alternative study designs focusing exclusively on CSE sentences, which may eliminate this problem.

Additionally, only 3SG subjects were used for copular constructions as the focus was on the effect of complement types, thus the results may not be generalisable to other types of subjects. Research by Ho and Platt (1993) suggests that the preceding environment does have an effect on copula use, with 1SG and 3SG pronouns promoting copula use, and other pronouns and NPs promoting a zero copula. The use of both 3SG pronominal and NP subjects in this study seems to cover both directions of influence of the preceding environment, although a broader study investigating the effects of other subjects may provide a more comprehensive description of copula optionality in CSE.

Finally, as mentioned above, the presentation of the stimuli via the visual modality may have limited their naturalness, given that CSE is much more often encountered aurally, and does not have a codified written form. This may have been particularly important for past tense marking, which is related to the phonetics of the past tense forms; indeed, it is plausible that CSE speakers have slightly different written and phonetic representations of past tense forms, leading to differential performance in the two modalities. While using auditory stimuli is inherently much more complicated, due

594 to having to control for aspects of the speaker's voice (e.g., perceived gender, accent), it
595 may elicit more accurate responses from respondents, and therefore permit finer-grained
596 descriptions of CSE.

597 Nonetheless, the results as a whole demonstrate substantial nuance in participants'
598 endorsement of SSE and CSE sentences. They also highlight the importance of
599 acceptability judgements as a tool to understand the syntax of a particular variety,
600 especially for cases with known variation across speakers, whether sociolinguistic or
601 otherwise. In conjunction with other approaches such as corpus and theoretical analyses,
602 such an empirical approach will allow for more accurate and detailed descriptions of
603 language as it is used by speakers, as well as a more systematic method of measuring
604 language change.

References

- Ansaldo, U. (2010). Contact and Asian Varieties of English. In *The Handbook of Language Contact* (pp. 498–517). John Wiley & Sons, Ltd.
<https://doi.org/10.1002/9781444318159.ch24>
- Arche, M. J., Fábregas, A., Marín, R., Arche, M. J., Fábregas, A., & Marín, R. (Eds.). (2019). Main questions in the study of copulas: Categories, structures, and operations. In *The Grammar of Copulas Across Languages* (pp. 1–30). Oxford University Press. <https://doi.org/10.1093/oso/9780198829850.003.0001>
- Arcodia, G. F. (2014). The Chinese adjective as a word class. In R. Simone & F. Masini (Eds.), *Word Classes: Nature, typology and representations* (pp. 95–118). John Benjamins Publishing Company. <https://doi.org/10.1075/cilt.332.06arc>
- Bennett, G. (2010). *Using Corpora in the Language Learning Classroom: Corpus Linguistics for Teachers*. Ann Arbor, MI: University of Michigan Press/ELT.
<https://doi.org/10.3998/mpub.371534>
- Bokhorst-Heng, W. D., Alsagoff, L., McKay, S., & Rubdy, R. (2007). English language ownership among Singaporean Malays: Going beyond the NS/NNS dichotomy. *World Englishes*, 26(4), 424–445. <https://doi.org/10.1111/j.1467-971X.2007.00521.x>
- Bokhorst-Heng, W. D., Rubdy, R., McKay, S. L., & Alsagoff, L. (2010). Whose English? Language Ownership in Singapore’s English Language Debates. In L. Lim, A. Pakir, & L. Wee (Eds.), *English in Singapore: Modernity and Management* (pp. 132–157). Hong Kong University Press.
<https://doi.org/10.5790/hongkong/9789888028436.003.0006>
- Chan, B. (2023). Afterthoughts and Right Dislocation in Colloquial Singapore English: An Experimental Approach. *Proceedings of the 46th Annual Penn Linguistics Conference*. Presented at the The 46th Annual Penn Linguistics Conference.
 Retrieved from <https://repository.upenn.edu/handle/20.500.14332/58737>
- Chang, Q. (2009). *Copula deletion in Colloquial Singapore English* (Master’s thesis,

- National University of Singapore). National University of Singapore, Singapore.
Retrieved from <https://scholarbank.nus.edu.sg/handle/10635/17710>
- Cruz-Ferreira, M. (2005). Past tense suffixes and other final plosives in Singapore English. *English in Singapore: Phonetic Research on a Corpus*. Retrieved from https://www.academia.edu/922719/Past_tense_suffixes_and_other_final_plosives_in_Singapore_English
- Department of Statistics, Ministry of Trade and Industry, Republic of Singapore. (2021). *Singapore Census of Population 2020, Statistical Release 1: Demographic Characteristics, Education, Language and Religion* [Data set]. Retrieved from <https://www.singstat.gov.sg/-/media/files/publications/cop2020/sr1/cop2020sr1.ashx>
- Deterding, D. (2003). Tenses and will/would in a corpus of Singapore English. In D. Deterding, E. L. Low, & A. Brown (Eds.), *English in Singapore: Research on Grammar* (pp. 31–38). Singapore: McGraw-Hill Education (Asia).
- Deterding, D. (2005). Emergent patterns in the vowels of Singapore English. *English World-Wide*, 26(2), 179–197. <https://doi.org/10.1075/eww.26.2.04det>
- Deterding, D. (2007). *Singapore English*. Edinburgh University Press. Retrieved from <https://www.jstor.org/stable/10.3366/j.ctt1r21h4>
- Djamouri, R., Paul, W., & Whitman, J. (2013). Postpositions vs Prepositions in Mandarin Chinese: The Articulation of Disharmony. In T. Biberauer & M. Sheehan (Eds.), *Theoretical Approaches to Disharmonic Word Order* (pp. 74–105). Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199684359.003.0003>
- Dowty, D. R. (1979). *Word Meaning and Montague Grammar* (G. Chierchia, P. Jacobson, & F. J. Pelletier, Eds.). Dordrecht: Springer Netherlands.
<https://doi.org/10.1007/978-94-009-9473-7>
- Fong, V. (2004). The verbal cluster. In L. Lim (Ed.), *Singapore English: A grammatical description* (pp. 75–104). John Benjamins Publishing Company.
<https://doi.org/10.1075/veaw.g33.06fon>

- 659 Foo, A. L., & Tan, Y.-Y. (2019). Linguistic insecurity and the linguistic ownership of
660 English among Singaporean Chinese. *World Englishes*, 38(4), 606–629.
661 <https://doi.org/10.1111/weng.12359>
- 662 Gupta, A. F. (1998). The situation of English in Singapore. In J. A. Foley, T. Kandiah,
663 Z. Bao, A. F. Gupta, L. Alsagoff, C. L. Ho, ... W. Bockhorst-Heng (Eds.), *English in*
664 *new cultural contexts: Reflections from Singapore* (pp. 106–126). Singapore: Oxford
665 University Press.
- 666 Gut, U. (2009). Past tense marking in Singapore English verbs. *English World-Wide*,
667 30(3), 262–277. <https://doi.org/10.1075/eww.30.3.02gut>
- 668 Higgins, C. (2003). “Ownership” of English in the Outer Circle: An Alternative to the
669 NS–NNS Dichotomy. *TESOL Quarterly*, 37(4), 615–644.
670 <https://doi.org/10.2307/3588215>
- 671 Ho, M.-L. (2003). Past tense marking in Singapore English. In D. Deterding, E. L. Low,
672 & A. Brown (Eds.), *English in Singapore: Research on Grammar* (pp. 39–47).
673 Singapore: McGraw-Hill.
- 674 Ho, M.-L., & Platt, J. T. (1993). *Dynamics of a Contact Continuum: Singaporean*
675 *English*. Oxford, New York: Oxford University Press.
- 676 Horsch, J. (2022). *The comparative correlative construction in World Englishes: A*
677 *usage-based construction grammar approach* (Dr. phil. thesis, Katholische Universität
678 Eichstätt-Ingolstadt). Katholische Universität Eichstätt-Ingolstadt.
679 <https://doi.org/10.17904/ku.opus-833>
- 680 I Wayan Arka. (2013). Nonverbal predicates in Austronesian and Papuan languages: An
681 LFG perspective. *Proceedings of the 6th International Seminar on Austronesian and*
682 *Non-Austronesian Languages and Literatures*, 30–46. Bali: Universitas Udayana.
683 Retrieved from [https://www.researchgate.net/publication/322831487_Nonverbal_](https://www.researchgate.net/publication/322831487_Nonverbal_predicates_in_Austronesian_and_Papuan_languages_an_LFG_perspective)
684 [predicates_in_Austronesian_and_Papuan_languages_an_LFG_perspective](https://www.researchgate.net/publication/322831487_Nonverbal_predicates_in_Austronesian_and_Papuan_languages_an_LFG_perspective)
- 685 Juzek, T. S. (2015). *Acceptability judgement tasks and grammatical theory* (DPhil thesis,

University of Oxford). University of Oxford, Oxford. Retrieved from

<https://ora.ox.ac.uk/objects/uuid:b276ec98-5f65-468b-b481-f3d9356d86a2>

Kalaivanan, K., Sumartono, F., & Tan, Y.-Y. (2020). The Homogenization of Ethnic

Differences in Singapore English? A Consonantal Production Study. *Language and*

Speech. <https://doi.org/10.1177/0023830920925510>

Khairul Taufiq Abu Bakar, & Nor Hashimah Jalaluddin. (2017). Kata pemerian ialah dan

adalah: Analisis pragmatik. *Jurnal Linguistik*, 21(1), 14–28.

Koh, J. X., Mislán, A., Khoo, K., Ang, B., Ang, W., Ng, C., & Tan, Y.-Y. (2019).

Building the Singapore English National Speech Corpus. *Interspeech 2019*, 321–325.

ISCA. <https://doi.org/10.21437/Interspeech.2019-1525>

Lai, E. Y., Tan, L., Wong, V., Loke, L. T. T., & Bond, F. (2013). The OPT-ional

Phenomenon in Singapore English: A Corpus-based Approach Using Time Annotated

Corpora. *Procedia—Social and Behavioral Sciences*, 95, 431–441.

<https://doi.org/10.1016/j.sbspro.2013.10.666>

Leimgruber, J. R. E. (2013). *Singapore English: Structure, variation, and usage*.

Cambridge: Cambridge University Press.

Leong, A. X. W. (2013). Colloquial Singapore English never. *Rice Working Papers in*

Linguistics, 4.

Leong, W. (2021). Sociolinguistic Factors Affecting Tense Variation in Singapore English.

Lifespans and Styles, 7(1), 36–47. <https://doi.org/10.2218/lis.v7i1.2021.5642>

Li, L. (2015). The V+V construction in Singaporean English. *Studies in the Linguistic*

Sciences: Illinois Working Papers, 40, 1–16. Retrieved from

<https://core.ac.uk/works/18873462>

Li, Y.-H. A. (1990). *Order and Constituency in Mandarin Chinese* (J. Maling & L. Rizzi,

Eds.). Dordrecht: Springer Netherlands. <https://doi.org/10.1007/978-94-009-1898-6>

Lieberman, E., Michel, J.-B., Jackson, J., Tang, T., & Nowak, M. A. (2007). Quantifying

the evolutionary dynamics of language. *Nature*, 449(7163), 713–716.

713 <https://doi.org/10.1038/nature06137>

714 Lim, L. (2004). Sounding Singaporean. In L. Lim (Ed.), *Singapore English: A*
715 *grammatical description* (pp. 19–56). John Benjamins Publishing Company.

716 <https://doi.org/10.1075/veaw.g33.04lim>

717 Lin, J., & Khoo, Y. K. (2018). Singapore Mandarin Chinese: Its variations and studies.
718 *Chinese Language and Discourse. An International and Interdisciplinary Journal*,
719 9(2), 109–135. <https://doi.org/10.1075/cld.18007.lin>

720 Liu, M. (2015). Tense and Aspect in Mandarin Chinese. In W. S.-Y. Wang & C. Sun
721 (Eds.), *The Oxford Handbook of Chinese Linguistics* (pp. 274–289). Oxford
722 University Press. <https://doi.org/10.1093/oxfordhb/9780199856336.013.0070>

723 McCawley, J. D. (1992). Justifying Part-of-Speech Assignments in Mandarin Chinese.
724 *Journal of Chinese Linguistics*, 20(2), 211–246. Retrieved from
725 <https://www.jstor.org/stable/23753906>

726 Moeljadi, D., Bond, F., & Costa, L. M. da. (2016). Basic copula clauses in Indonesian.
727 *Proceedings of the International Conference on Head-Driven Phrase Structure*
728 *Grammar*, 442–456. <https://doi.org/10.21248/hpsg.2016.23>

729 Mustaffa, A. R. (2018). Silent Syntactic Structures in Malay: Copular Clauses. *Journal*
730 *of Modern Languages*, 28, 20–43. Retrieved from
731 <https://jml.um.edu.my/index.php/JML/article/view/13172>

732 Nik Safiah Karim, Farid M. Onn, Hashim Haji Musa, & Abdul Hamid Mahmood. (1989).
733 *Tatabahasa dewan*. Kuala Lumpur: Dewan Bahasa dan Pustaka.

734 Olsen, M. (1994). The semantics and pragmatics of lexical aspect features. *Studies in the*
735 *Linguistic Sciences*, 24, 361–375.

736 Pakir, A. (1999). Bilingual education with English as an official language: Sociocultural
737 implications. In J. E. Alatis & A.-H. Tan (Eds.), *Georgetown University Round Table*
738 *on Language and Linguistics, 1999: Language in Our Time* (pp. 11–30). Washington,
739 DC: Georgetown University Press.

- Paul, W. (2010). The rehabilitation of a much ostracized category: Adjectives in Mandarin Chinese. In P. Cabredo Hofherr & O. Matushansky (Eds.), *Adjectives: Formal analyses in syntax and semantics* (pp. 115–152). John Benjamins Publishing Company. <https://doi.org/10.1075/la.153.04pau>
- Platt, J. T. (1975). The Singapore English speech continuum and its basilect “Singlish” as a “creoloid.” *Anthropological Linguistics*, 17(7), 363–374.
- Platt, J. T. (1979). Variation and implicational relationships: Copula realization in Singapore English. *General Linguistics*, 19(1), 1–14. Retrieved from <https://www.proquest.com/docview/1301501321>
- Ross, C. (1991). Coverbs and Category Distinctions in Mandarin Chinese. *Journal of Chinese Linguistics*, 19(1), 79–115. Retrieved from <https://www.jstor.org/stable/23757091>
- Rubdy, R., McKay, S. L., Alsagoff, L., & Bokhorst-Heng, W. D. (2008). Enacting English language ownership in the Outer Circle: A study of Singaporean Indians’ orientations to English norms. *World Englishes*, 27(1), 40–67. <https://doi.org/10.1111/j.1467-971X.2008.00535.x>
- Sato, Y. (2013). Wh-Questions in Colloquial Singapore English: Adaptive traits from vernacular Malay and typological congruence. *Journal of Pidgin and Creole Languages*, 28(2), 299–322. <https://doi.org/10.1075/jpcl.28.2.02sat>
- Schneider, E. W. (2007). *Postcolonial English: Varieties around the World*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511618901>
- Sharma, D., & Rickford, J. R. (2009). AAVE/creole copula absence: A critique of the imperfect learning hypothesis. *Journal of Pidgin and Creole Languages*, 24(1), 53–90. <https://doi.org/10.1075/jpcl.24.1.03sha>
- Tan, A. W. M. (2020). *Copular constructions in Colloquial Singaporean English* (Undergraduate linguistics project, University of Oxford). University of Oxford, Oxford. Retrieved from https://lagb.org.uk/resources/Documents/AlvinTan_diss.pdf

- Tang, S.-W. (2001). Nominal predication and focus anchoring. *ZAS Papers in Linguistics*, 22, 159–172. <https://doi.org/10.21248/zaspil.22.2001.105>
- Terassa, L. A. M. (2019). *Morphological Simplification in Asian Englishes: Frequency, Substratum Transfer, and Institutionalization* (Dr. phil. thesis, Universität Freiburg). Universität Freiburg, Freiburg im Breisgau. Retrieved from <https://books.google.com?id=gqzNzQEACAAJ>
- Vendler, Z. (1957). Verbs and Times. *The Philosophical Review*, 66(2), 143–160. <https://doi.org/10.2307/2182371>
- Winstedt, R. O., & Wilkinson, R. J. (1913). *Malay grammar*. Oxford: Clarendon Press. Retrieved from <http://archive.org/details/malaygrammar00winsrich>
- Wong, B. E. (2012). Acquisition of English tense and agreement morphology by L1 Malay and L1 Chinese speakers. *3L: Language, Linguistics, Literature—The Southeast Asian Journal of English Language Studies*, 18(3), 5–14. Retrieved from <http://www.ukm.my/ppbl/3L/3LHome.html>
- Woon, F. T. (2018). *Linguistic sound symbolism and reading development : Sound-shape matching and predictors of reading in multilingual Singapore* (Master's thesis). <https://doi.org/10.32657/10356/73469>
- Wu, C.-Y., O'Brien, B. A., Styles, S. J., & Chen, S.-H. A. (2020). The Impact of Bilingualism on Skills Development and Education. In S. C. Tan & S.-H. A. Chen (Eds.), *Transforming Teaching and Learning in Higher Education: A Chronicle of Research and Development in a Singaporean Context* (pp. 47–69). Singapore: Springer. https://doi.org/10.1007/978-981-15-4980-9_3
- Xu, J. (2003). Focus-Marking in Chinese and Malay : A Comparative Perspective. In D. H. Ji & K. T. Lua (Eds.), *Proceedings of the 17th Pacific Asia Conference on Language, Information and Computation* (pp. 2–15). Sentosa, Singapore: COLIPS Publications. <https://doi.org/http://hdl.handle.net/2065/12272>
- Yu, J. (2015). *Copula-less, Non-verbal Predication in Colloquial Singapore English and*

794 *the General Anchoring Condition* (Master's thesis, National University of Singapore).

795 National University of Singapore, Singapore. Retrieved from

796 <https://scholarbank.nus.edu.sg/handle/10635/119819>

797 Zhu, X. (2016). 新加坡华语语法变异研究 [*A study of grammatical variation in Singapore*

798 *Mandarin*]. 世界图书出版公司北京公司. Retrieved from

799 <https://books.google.com?id=pW8QkAEACAAJ>

Appendix: Questionnaire

This appendix contains the full list of sentences used in the acceptability judgement task, the rating scale, as well as the demographic questions presented to participants.

Acceptability judgement items

The items used in the acceptability judgement task are shown in Table 2. Words in braces indicate alternations between SSE versions (before the slash) and CSE versions (after the slash) of the sentence.

Table 2

Items for acceptability judgement task.

No.	Type	Sentence
A.	Practice – CSE	He cannot anyhow say people.
B.	Practice – CSE	She go home already.
C.	Practice – CSE	I don't have ticket.
D.	Practice – CSE	They got time to finish one.
E.	Practice – CSE	We sit here talk.
F.	Practice – SSE	Her daughter is very friendly.
G.	Practice – SSE	They will wipe the table.
H.	Practice – SSE	My uncle has finished eating.
I.	Practice – SSE	The fuel in the tank is enough.
J.	Practice – SSE	The rabbit has too much food.
1.	Copula – NP	John {is / $\emptyset$ } a doctor.
2.	Copula – NP	Lucy {is / $\emptyset$ } the English teacher.
3.	Copula – NP	My friend {is / $\emptyset$ } a chef.
4.	Copula – NP	His cousin {is / $\emptyset$ } the older one.

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| 5. | Copula – NP | The apple {is / $\emptyset$ } one dollar. |
| 6. | Copula – NP | Now {is / $\emptyset$ } two o'clock. |
| 7. | Copula – NP | Sally {is / $\emptyset$ } a baker. |
| 8. | Copula – NP | James {is / $\emptyset$ } the car mechanic. |
| 9. | Copula – NP | His wife {is / $\emptyset$ } an actress. |
| 10. | Copula – NP | Her boss {is / $\emptyset$ } the taller one. |
| 11. | Copula – NP | That one {is / $\emptyset$ } eighty cents. |
| 12. | Copula – NP | Tomorrow {is / $\emptyset$ } Saturday. |
| 13. | Copula – AP | Marcus {is / $\emptyset$ } good. |
| 14. | Copula – AP | Ann {is / $\emptyset$ } clever. |
| 15. | Copula – AP | The kid {is / $\emptyset$ } careless. |
| 16. | Copula – AP | The camp {is / $\emptyset$ } tiring. |
| 17. | Copula – AP | Their shop {is / $\emptyset$ } far. |
| 18. | Copula – AP | Your dog {is / $\emptyset$ } big. |
| 19. | Copula – AP | Samuel {is / $\emptyset$ } happy. |
| 20. | Copula – AP | Kate {is / $\emptyset$ } pretty. |
| 21. | Copula – AP | Their uncle {is / $\emptyset$ } old. |
| 22. | Copula – AP | The bus {is / $\emptyset$ } fast. |
| 23. | Copula – AP | The supermarket {is / $\emptyset$ } open. |
| 24. | Copula – AP | Your flat {is / $\emptyset$ } nice. |
| 25. | Copula – PP | Tom {is / $\emptyset$ } at home. |
| 26. | Copula – PP | The meeting {is / $\emptyset$ } at noon. |
| 27. | Copula – PP | Her toy {is / $\emptyset$ } in the box. |
| 28. | Copula – PP | Breakfast {is / $\emptyset$ } in the morning. |
| 29. | Copula – PP | Your book {is / $\emptyset$ } on the table. |
| 30. | Copula – PP | Our trip {is / $\emptyset$ } on Tuesday. |
| 31. | Copula – PP | Mary {is / $\emptyset$ } at work. |

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| 32. | Copula – PP | Our flight {is / $\emptyset$ } at seven. |
| 33. | Copula – PP | Food {is / $\emptyset$ } in the kitchen. |
| 34. | Copula – PP | The show {is / $\emptyset$ } in twenty minutes. |
| 35. | Copula – PP | My robot {is / $\emptyset$ } on the floor. |
| 36. | Copula – PP | Their wedding {is / $\emptyset$ } on a weekend. |
| 37. | Past – Suppletion | Yesterday I {went / go} there. |
| 38. | Past – Suppletion | In the sixties he {was / is} a policeman. |
| 39. | Past – Suppletion | Ten years ago, the houses {were / are} very big. |
| 40. | Past – Suppletion | The exam last Tuesday {was / is} very hard. |
| 41. | Past – Suppletion | Just now they {went / go} to the shop. |
| 42. | Past – Suppletion | You thought your bags {were / are} lost. |
| 43. | Past – Vowel | She said that you {took / take} her book. |
| 44. | Past – Vowel | My sister {found / find} a job two days ago. |
| 45. | Past – Vowel | Last Monday they {came / come} twice. |
| 46. | Past – Vowel | She {ate / eat} the cake last night. |
| 47. | Past – Vowel | I {met / meet} your sister at the wedding last year. |
| 48. | Past – Vowel | Mark {dug / dig} a hole so he can plant vegetables now. |
| 49. | Past – [t]/[d] | Before this he {stayed / stay} at his friend's place. |
| 50. | Past – [t]/[d] | I {tried / try} to do it myself last week. |
| 51. | Past – [t]/[d] | Henry {packed / pack} for his trip this morning. |
| 52. | Past – [t]/[d] | He {brushed / brush} his hair when he woke up. |
| 53. | Past – [t]/[d] | My brother {walked / walk} the dog yesterday. |
| 54. | Past – [t]/[d] | Jane {pushed / push} me first. |
| 55. | Past – [ɪd] | When I was young, I {wanted / want} to be a doctor. |
| 56. | Past – [ɪd] | Last time Sarah {hated / hate} carrots. |
| 57. | Past – [ɪd] | This blue shirt {faded / fade} in the wash. |
| 58. | Past – [ɪd] | His mother {painted / paint} the wall white the first time. |

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| 59. | Past – [ɪd] | They {ended / end school} last week. |
| 60. | Past – [ɪd] | I {folded / fold} the clothes just now. |
| 61. | Past – Vwl + [t]/[d] | His wife {brought / bring} cake for the last party. |
| 62. | Past – Vwl + [t]/[d] | Your friend {told / tell} me what happened. |
| 63. | Past – Vwl + [t]/[d] | They {sold / sell} their house last month. |
| 64. | Past – Vwl + [t]/[d] | Their father {bought / buy} the house three years ago. |
| 65. | Past – Vwl + [t]/[d] | Her cousin {taught / teach} primary school in the past. |
| 66. | Past – Vwl + [t]/[d] | The food {felt / feel} very cold before I put it in the oven. |
| 67. | Benchmark – Low | You become in the house. |
| 68. | Benchmark – Low | She become at the office. |
| 69. | Benchmark – Low | The box become on the floor. |
| 70. | Benchmark – High | He became a teacher. |
| 71. | Benchmark – High | Their cousin became skinnier. |
| 72. | Benchmark – High | The sky became dark. |
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807 **Rating scale**

808 The scale that participants were asked to use for ratings is shown in Table 3.

Table 3

Rating scale for acceptability judgement task.

Rating	Description
1	This isn't a good sentence. It's impossible to understand. I would not use this sentence. No speaker from Singapore would use this sentence.
2	This sentence is better than one I'd rate as 1, but it's still very bad. It's very difficult to understand. Most people from Singapore would definitely not say this.
3	This sentence is not good enough to be rated 4, but it's better than a 2. It's difficult to understand. Somebody from Singapore might use it, but I'd be unlikely to use it myself.
4	This sentence is less than perfect. It sounds a bit strange to me. Other people from Singapore might use it, but I'm not totally comfortable with it.
5	This sentence sounds fine. I'd use it without hesitation, and so would other people from Singapore.

809 **Demographic questions**

810 The demographic questions presented to participants are shown in Table 4. Some of
 811 the questions required participants to select an option; in these cases, the possible
 812 options are listed in brackets.

Table 4

Demographic questions.

No.	Question
1.	Age [18, 19, 20, 21, 22, 23, 24, 25]
2.	Gender [Male, Female, Other, Prefer not to say]
3.	Which university are you attending / have you attended / are you about to attend? (Type “NA” if you are not intending to attend university)
4.	Which junior college / polytechnic / centralised institute did you attend?
5.	What is your mother tongue?
6.	How would you rate your ability in your mother tongue? (1: limited comprehension, 5: fluent and can communicate easily) [1, 2, 3, 4, 5]
7.	How many years of formal learning have you had in your mother tongue?
8.	Do you speak any other languages / dialects? If so, which ones?
9.	How would you rate your ability in these other languages / dialects? (1: limited comprehension, 5: fluent and can communicate easily) [1, 2, 3, 4, 5]
10.	How many years of formal learning have you had in these other languages / dialects?
11.	Were you born in Singapore? [Yes, No]
12.	How long have you lived / did you live in Singapore?
13.	Have you lived in any other countries? If so, when and for how long?